B.TECH IT • MINOR PROJECT

AQUA

AI-based Quality Understanding and Analytics

A machine-learning and full-stack framework for water-quality forecasting and early warning

Minor Project Presentation | October 2026

GTBIT • Guru Gobind Singh Indraprastha University

AQUA at a glance

Data scale and headline results from the reported experiments

12,062

station-date records in the Ganga master table

95

stations, 2021–2025 (through 23 Apr 2025)

1,724

matched Yamuna rows across 15 stations

0.5267

Yamuna BOD test R^2 (2024, N = 204), RMSE 9.5471 vs 11.2376 for persistence

0.2048

Ganga BOD test R^2 for Extra Trees, vs −0.0091 for persistence (N = 117)

Useful but modest signal — and persistence remains a strong baseline that every result is judged against.

Why AQUA?

The gap between periodic sampling and timely decisions

Periodic observations

Water-quality samples are collected intermittently—not as a continuous stream. Deterioration can occur between sampling visits.

Data is fragmented

Chemical, physical and biological measurements may have different dates, missing values and station naming conventions.

Prediction is not trivial

Sparse, irregular records and station-to-station variation make reliable forecasting and generalization difficult.

PROJECT QUESTION

Can historical water-quality patterns support useful forecasts and evidence-backed early warnings—while making uncertainty and model limitations visible?

AQUA is a decision-support framework, not a replacement for laboratory sampling or regulatory assessment.

The AQUA approach

From monitoring data to a forecast that can be inspected and acted upon

01

Collect

CPCB / CWC water-quality records; weather where valid

02

Prepare

Clean timestamps, align stations, check coverage

03

Engineer

Station history, lags, calendar and other features

04

Forecast

Target-specific regression models + persistence check

05

Interpret

Trends, errors, potential warnings and model context

IMPLEMENTED / EXPERIMENTAL

Dataset preparation, temporal ML experiments, baseline comparisons and diagnostic analyses.

PLANNED APPLICATION LAYER

Integrate forecasts into a usable dashboard with station views, clear caveats and alert logic.

Scope, data and prediction targets

The current modelling work uses two river pipelines; coverage differs by experiment

Yamuna • BOD

Station-aware surface-water observations. Reported chronological experiment: train 2021–22, validate 2023, test 2024. Matched dataset previously recorded 1,724 rows across 15 stations.

Ganga • BOD, DO, pH

Uttar Pradesh CWC surface-water data. Chemical, biological and physical files merged by station and acquisition timestamp; master: 12,062 station-date records, 95 stations, 2021–2025 (through 23 Apr 2025).

PREDICTION TARGETS

BOD	Organic pollution indicator
DO	Oxygen available in water
pH	Acidity / alkalinity

Weather and discharge are candidate contextual inputs only where temporal/spatial compatibility is demonstrated; no positive weather contribution is claimed from the available experiments.

Methodology and evaluation design

Forecast the next observed value; preserve time order to reduce leakage

Preprocessing

Parse acquisition timestamps; standardize station keys; merge by station + time; inspect missingness and invalid values; use training-only median imputation for model features.

Feature engineering

Use station-specific history and lagged observations, plus eligible water-quality/context features. Lags encode recent conditions; they must be shifted so future target values cannot leak.

Evaluation

MAE and RMSE quantify error; R^2 describes variance explained relative to a mean predictor. Compare with persistence: predict the last observed value.

CHRONOLOGICAL SPLITS

Train › Validate › Test

Earlier years fit the model; a later period supports model selection; the final held-out period estimates forward-in-time performance. Exact split details vary by river experiment.

No random row split is used for the reported chronological evaluations. The exact final estimator settings should be checked against the executed notebook.

Progress 1 | Data collection and preparation

A usable master table is a modelling result in its own right

CWC / Ganga

Chemical • biological • physical files

95 stations | 2021–2025

CPCB / Yamuna

Surface-water records

1,724 matched rows | 15 stations

Context data

Weather matching feasibility tested; not assumed beneficial by default

NORMALIZE → VALIDATE → MERGE ON STATION + ACQUISITION TIME

Timestamp checks

Parse source date format; inspect chronology and gaps.

Quality checks

Invalid pH values >14 set missing; non-positive pH excluded from pH target.

Model-ready data

Remove missing target rows; impute feature gaps from training medians.

Key data finding: monitoring is irregular (Ganga median observation gap ≈10 days; maximum reported gap ≈323 days).

Progress 2 | Yamuna BOD forecasting

Gradient Boosting Regressor • chronological evaluation • supplied result set

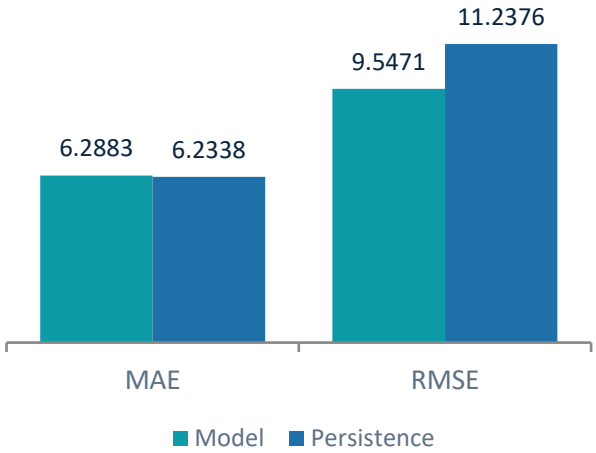
Period	Rows	Model MAE	Model RMSE	Model R^2
Train 2021–2022	678	—	—	—
Validation 2023	470	8.3442	14.7699	0.3375
Test 2024	**204**	**6.2883**	**9.5471**	**0.5267**

Validation

Model improves over persistence on all three reported metrics.

Test nuance

MAE is marginally worse than persistence (6.2883 vs 6.2338), but RMSE and R^2 improve.



Interpretation: useful signal, but high BOD values are underpredicted. Error tails and station selection need further validation.

Progress 3 | Ganga multi-target results

Target-specific estimators; reported test set N = 117 for each model

Target	Estimator	Model MAE	Model RMSE	Model R^2	Persistence MAE / RMSE / R^2
BOD	Extra Trees	1.7539	2.3435	0.2048	1.9282 / 2.6401 / −0.0091
DO	Extra Trees	0.6376	0.8726	0.2481	0.6207 / 0.8551 / 0.2838*
pH	HistGradientBoosting	0.1101	0.1721	**−0.2033**	0.1291 / 0.2206 / −0.9787

BOD — Beats persistence

Beats persistence on MAE, RMSE and R^2; explanatory strength remains modest.

DO — Persistence stronger

Does not beat persistence on the reported test metrics; baseline N=116.

pH — Mixed result

Lower errors than persistence, but negative R^2 means weak variance explanation.

**Persistence DO: N=116. Metrics are target-specific; do not compare raw errors across parameters with different scales/units.*

Progress 4 | What the metrics actually say

A baseline comparison prevents us from mistaking "better than nothing" for useful forecasting

MAE

Average absolute error. Easy to interpret in the target's units; less sensitive to outliers than RMSE.

RMSE

Penalizes large misses more strongly; useful when occasional large errors matter.

R^2

Can be negative. Negative R^2 means worse squared-error performance than predicting the test-set mean.

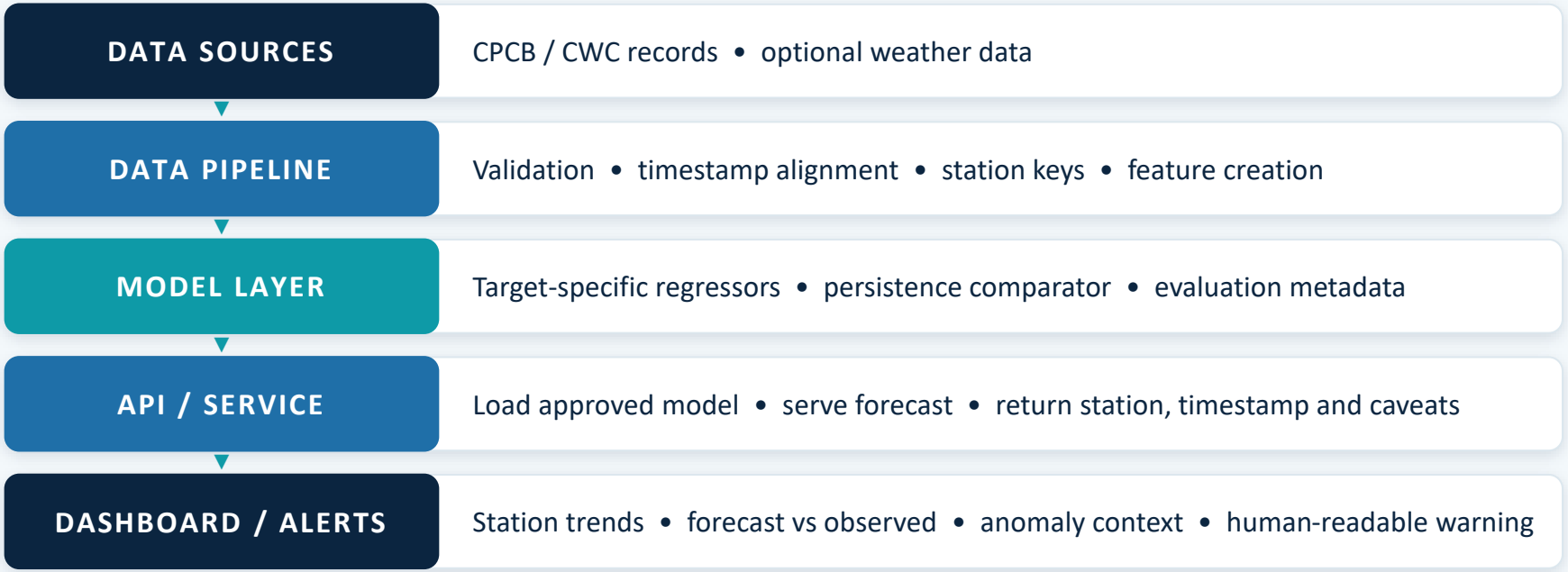
PERSISTENCE **Predict the latest available observed value. This is a strong sanity check when water quality changes slowly.**

DIAGNOSTIC TAKEAWAYS

- Yamuna: high BOD observations are underpredicted; MAE alone hides this tail risk.
- Ganga DO: persistence remains the stronger test comparator.
- Ganga pH: lower MAE/RMSE do not cancel the negative R^2; calibration and target variation need inspection.

Proposed end-to-end system architecture

ML pipeline is experimental; application integration remains planned work



Planned safeguards: model/version logging, stale-data checks, uncertainty/context display and human review of alerts.

Limitations and research considerations

A credible result includes the conditions under which it may fail

Irregular sampling

A dominant ~10-day cadence does not mean continuous monitoring; long gaps remain.

Station-selection validity

Confirm that station filters represent the intended river system and are not mixing incompatible sites.

High-value underprediction

Yamuna BOD errors rise on high observations; averages can conceal consequential misses.

Mixed target performance

Ganga DO does not beat persistence; pH has negative R^2 despite lower absolute errors.

Weather / hydrology

No positive weather contribution should be claimed without a validated ablation result.

Operational readiness

A forecast is not a regulatory classification; dashboard and alert integration are not claimed complete.

Use cautious language: “potential risk signal” rather than “confirmed contamination.”

Next steps

From model outputs to a usable decision-support tool

1. **Verify the frozen notebook**
   Confirm estimator parameters, feature columns, split logic and saved metrics.

2. **Strengthen diagnostics**
   Review high-BOD errors, residuals, station-level scores and baseline gaps.

3. **Integrate carefully**
   Expose forecast + timestamp + data freshness + model version through an API.

4. **Design cautious alerts**
   Use evidence-backed thresholds, communicate uncertainty and retain human oversight.

AQUA's current contribution: a time-aware ML evaluation framework with honest baseline comparisons—and a clear path toward explainable early warning.



Thank you

Questions & discussion

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